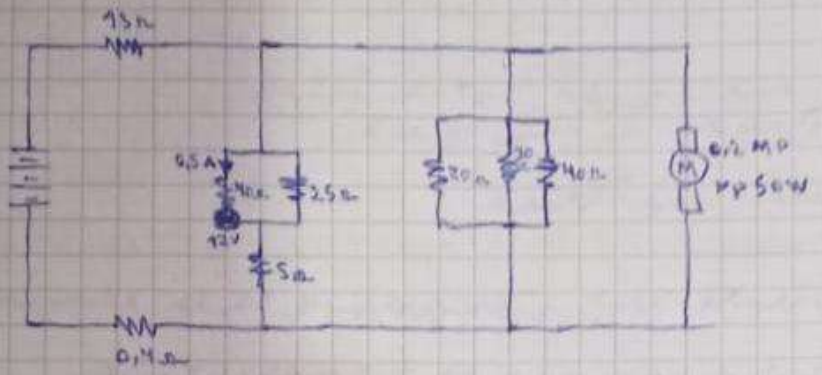
CIRCUITOS ELECTRICOS Problema 4 y 5 realizado

Problema 5 T = 45 DIAS > 7800 hr (1HP > 746 W)



CALCULAR
$P_L = 6W$
$\eta = 75\%$
$U = 272,53V$
$I_T = 77,08A$
$P = 2\,343,73W$
$E = 4218,78 Kwh$

LAMPARA

$I_L = 0,5A$
$V_L = 12V$
$R_L = 24\Omega$

$R_L = \frac{V_L}{I_L} = \frac{12V}{0,5A} = 24\Omega$

MOTOR

$P_U = 0,2 HP \times 746W = 149,2W$
$P_P = 50W$

RAMA I -> resistencia

R_1 R_L
$40\Omega + 24\Omega = 64\Omega$

$V_{paralelo} = 0,5A \times 64\Omega = 32V$

$I_a = \frac{32V}{25\Omega} = 1,28A$

$I_R = I_L + I_R = 0,5A + 1,28A = 1,78A$

$U_{panel} = V_{para} + (I_R \times 5\Omega) \Rightarrow 32V + (1,78A \times 5\Omega) = 32V + 8,9V = 40,9V$

RAMA -> todas las resistencias

$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{30} + \frac{1}{40} = \frac{6+4+3}{120} = \frac{13}{120} \rightarrow 9,23\Omega$

$I = \frac{V_{para}}{R_T} = \frac{40,9V}{9,23\Omega} = 4,43A$

RAMA MOTOR

$I_M = \frac{P_{Mr}}{U_{rama}} = \frac{199,2W}{40,9V} = 4,87A$

$I_T = I_{RAMA1} + I_{resistencia} = I_M = 1,78A + 4,13A + 4,87A = 11,08A$

$U = U_{rama_1} + I_T \times (15\Omega + 0,4\Omega)$

$U = 40,9V + 11,08A \times 15,4\Omega = 40,9V + 170,63V = 211,53V$

CALCULOS

$P_L = V_L \times I_L = 12V \times 0,5A = 6W$

$P = U \times I_T = 211,53V \times 11,08A = 2343,72W$
$2,35kW$

$\eta = \frac{P_U}{P_{ES}} \times 100 = \frac{149,2W}{199,2W} \times 100 = 74,9\% \sim$

$U = 211,53V$

$I_T = 11,08A$

$E = P \times T = 2,34375KW \times 1800h = 4218,75KWh$